1.02 Surds

Chapter 1 · Original slide 22

5: Surds

22

Recap:

A surd is a root of a number that does not simplify to a rational number.

Laws:

$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$

$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$

Note: A *rational* number is any which can be expressed as $\frac{a}{b}$ where a, b are integers. $\frac{2}{3}$ and $\frac{4}{1} = 4$ are rational numbers, but π and $\sqrt{2}$ are not.

$\sqrt{3} \times 2$

$3\sqrt{5} \times 2\sqrt{5}$

$\sqrt{8} = \sqrt{4}\sqrt{2}$

$\sqrt{12} + \sqrt{27}$

$\left(\sqrt{8} + 1\right)\left(\sqrt{2} - 3\right)$

Chapter 1 · Original slide 23

Exercise 1E – Pure Y1 Page 13

23

3 Simplify $\sqrt{75} - \sqrt{12}$ giving your answer in the form $a\sqrt{3}$, where a is an integer.

Extension

1 [SMC 2014 Q24] Which of the following is smallest?
$10 - 3\sqrt{11}$; $8 - 3\sqrt{7}$; $5 - 2\sqrt{6}$; $9 - 4\sqrt{5}$; $7 - 4\sqrt{3}$.

2 [SMC 2012 Q21] Which of the following numbers does not have a square root in the form $x + y\sqrt{2}$, where x and y are positive integers?
$17 + 12\sqrt{2}$; $22 + 12\sqrt{2}$; $38 + 12\sqrt{2}$; $54 + 12\sqrt{2}$; $73 + 12\sqrt{2}$.

1.02 Surds

Chapter 1 · Original slide 24

6: Rationalising The Denominator

24

Here's a surd. What could we multiply it by such that it's no longer an irrational number?

$$\sqrt{5} \times \square = \square$$

Chapter 1 · Original slide 25

Examples

25

$$\square \quad \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$$

$$\frac{6}{\square} = 2\sqrt{3}$$

$$\frac{7}{\sqrt{7}} = \frac{7\sqrt{7}}{7} = \sqrt{7}$$

$$\frac{15}{\sqrt{5}} \quad \square \quad + \sqrt{5} = 4\sqrt{5}$$

1.02 Surds

Chapter 1 · Original slide 26

More Complex Denominators

26

You've seen 'rationalising a denominator', the idea being that we don't like to divide things by an irrational number.

But what do we multiply the top and bottom by if we have a more complicated denominator?

$$\frac{1}{\sqrt{2}+1} \times \square = \square$$

Chapter 1 · Original slide 27

More Examples

27

$$\frac{3}{\sqrt{6}-2} \times \square = \square$$

$$\frac{4}{\sqrt{3}+1} \times \square = \square \quad 2\square$$

$$\frac{3\sqrt{2}+4}{5\sqrt{2}-7} \times \frac{5\sqrt{2}+}{5\sqrt{2}+} \square \sqrt{} \square$$

1.02 Surds

Chapter 1 · Original slide 28

Test Your Understanding

28

Rationalise the denominator and simplify $\frac{4}{\sqrt{5}-2}$

Rationalise the denominator and simplify $\frac{2\sqrt{3}-1}{3\sqrt{3}+1}$

$\frac{2\sqrt{3}}{3\sqrt{3}}$
$= \frac{18}{}$
$= \frac{19}{}$

AQA IGCSE FM June 2013 Paper 1

Solve $y\left(\sqrt{3}-1\right)=8$

Give your answer in the form $a+b\sqrt{3}$ where a and b are integers.

$\mathbf{y}=$

Chapter 1 · Original slide 29

Writing surd expressions in power form

29

This is not in the textbook, but a common type of question in exams, often asked in conjunction with differentiation (a later chapter).

Express $\frac{(2+\sqrt{x})^2}{x}$ as powers of x.

Test Your Understanding

Express $\frac{(x+\sqrt{x})^2}{2\sqrt{x}}$ as powers of x.

1.02 Surds

Chapter 1 · Original slide 30

Exercise 1F (Page 15)

30

or alternatively: (not in textbook)

1 Rationalise the denominator and simplify the following:

$\frac{1}{\sqrt{5}+2} =$

a $\frac{\sqrt{3}}{\sqrt{3}-1} = \frac{3+\sqrt{3}}{2}$

b $\frac{\sqrt{3}+1}{\sqrt{5}-2} = 7+3\sqrt{}$

c $\frac{}{3\sqrt{3}+4} = 2-\sqrt{}$

d $\frac{5\sqrt{5}-2}{}{-3} = 4+\sqrt{}$

e

2 Expand and simplify:

$\left(\sqrt{5}+3\right)\left(\sqrt{5}-2\right)\left(\sqrt{5}+1\right)$

3 Rationalise the denominator, giving your answer in the form $a+b\sqrt{3}$.

$\frac{3\sqrt{3}+7}{3\sqrt{3}-5} =$

4 Solve $x\left(4-\sqrt{6}\right) = 10$ giving your answer in the form $a+b\sqrt{6}$.

$x = \frac{10}{} = 4+\sqrt{6}$

Solv

5 $y = \frac{3+\sqrt{2}}{1+\sqrt{2}} = 2\sqrt{2}-1$

Simplify:

$\frac{\sqrt{a+1}-\sqrt{a}}{\sqrt{a+1}+\sqrt{a}}$ 1

6